Chapter 7: Privacy Utility trade off in Synthetic and anonymized datasets

7.1 Concept : The main asked here is :

Does synthetic data really offer a better privacy–utility trade-off than traditional anonymization such as k-anonymity?

Suppose a Hospital has a dataset with Patient Name, age, sex, diagnosis, treatment and outcome.
They want to publish it for the purpose of research, but more research means more information and more information means more privacy problems.
This is termed as a privacy utility trade off .

Two approaches could be : Anonymized dataset , and the other is creating a synthetic dataset, by learning the relationships.

7.2 Why?

Synthetic data are often described as safer because they contain **newly generated records rather than direct copies of real patients**.But this does **not automatically mean they are private**.

7.3 Privacy loss
Whether an attacker can **learn something about an individual from the released synthetic data or generation mechanism that they could not reasonably learn otherwise.**

There are several possible privacy threats.

A. Re-identification: Can an attacker determine which real person corresponds to a synthetic/released record?

B. Membership inference: Can an attacker determine whether a particular person's data were used to train the generator?
"Was this person in the training data?"

C. Attribute inference: Can an attacker infer a sensitive characteristic?
For example:"Does this patient have cancer?"

D. Linkage attacks: Can the attacker combine the released data with another external dataset to learn something about an individual?

These are different threats, so **privacy must always be evaluated relative to a threat model**.

7.4 Privacy lost or not?

Suppose: D real→G→D syn

An attacker receives D syn.They also have some background information.
They then attempt an attack.If the attacker can reliably distinguish something about a real individual, the synthetic-data release has a privacy problem.

7.5 Membership Inference

Suppose we have: D train used to train the synthetic-data generator.

And: D nontrain which contains people who were not used for training.

We give the attacker a target record x.

The attacker has to predict:

b={1, x ∈ Dtrain}
b= {0, x ∈ D nontrain}

So the attack asks:**Can the attacker distinguish members from non-members?**

If the attacker performs no better than random guessing, that's good for privacy.
If the attacker performs very well, that's evidence of privacy leakage.

7.6 Measuring attack

Attacker advantage : TPR - FPR ,
Where TPR = true positive rate, FPR = false positive rate

7.7 What causes privacy loss :

Reasons:
1. Overfitting
2.Memorizing
3.Rare records

Note : Privacy–utility evaluation of synthetic data concerns the balance between limiting the information that can be learned about individuals in the original data and preserving the information and relationships necessary for the intended analytical task.

7.7 Fundamental Tension

Fundamental tension is:**The more accurately a model learns the original data, the greater the possibility that it learns information specific to individual records.**

7.7.1 DCR | NNDR | IMS

DCR – Distance to Closest Record

- DCR measures how close each synthetic record is to its nearest record in the training dataset.
- For each synthetic record, we find the minimum distance to any training record.
- A higher DCR means that, on average, synthetic records are farther from the training records and is therefore assumed to indicate better privacy protection.
- A lower DCR means that synthetic records are closer to training records and may indicate a higher privacy risk.

How to calculate DCR:

1. Start with the original dataset.
2. Divide the original dataset into:
   - Training / members – used to train the synthetic-data generator.
   - Test / holdout / non-members – not used to train the generator.
3. Generate the synthetic dataset using only the training data.
4. Determine the type of variables:
   - If all variables are continuous, a distance such as Euclidean distance can be used.
   - If the dataset contains both continuous and categorical variables, a distance such as Gower distance can be used.
5. For each synthetic record, calculate its distance to all training records.
6. Take the minimum distance as that synthetic record's DCR.
7. The continuous DCR measure can then be calculated as the mean of these minimum distances.

7.7.2 NNDR(Nearest neighbor distance ration)

For each record in the synthetic dataset, calculate the ratio between its distance to the nearest training record and its distance to the second-nearest training record.

$$NNDR_i = \frac{d(s_i, x_{(1)})}{d(s_i, x_{(2)})}$$

where:

- **x(1) = closest training record**
- **x(2) = second-closest training record.**

7.7.3 IMR (Identical match Share)

IMS defines a scalar distance measure as the number of records in [the synthetic dataset] with an identical match in [the training dataset].

7.7.4 DCR|NNDR|IMS as proxy

If synthetic records are much closer to training records than to holdout records, that could indicate memorization/privacy leakage.

If the distances are similar—or synthetic data is not unusually close to training data—there is less evidence of this type of privacy leakage.

7.7.5DCR privacy test: We have initially calculated the DCR from train to syn.
Now we need to calculate DCR from test to syn .
And compares the **5th percentiles** of these distance distributions.

Conceptually: Q0.05(DCRtrain) versus Q0.05(DCRholdout)

Why the 5th percentile?

The 5th percentile focuses on the **closest/suspicious synthetic records, than average.**

If synthetic DCR closer to DCR train, then it indicates privacy problem.

7.7.6 NNDR privacy test

The same basic idea is applied to NNDR.
We compare the NNDR train to NNDR test.

Then compare their **5th percentiles**, according to the reference paper.

7.7.7 IMS privacy test
IMS train < IMSholdout

7.7.8 Interpreting DCR, NNDR and IMS

DCR, NNDR and IMS are proxy measures of privacy risk. They do not directly prove that an individual can be identified or that membership can be inferred.

The main idea is to compare the synthetic data with both training members and non-members.

- If synthetic records are unusually close to training records, this may indicate memorization or overfitting.
- If synthetic records are similarly distant from members and non-members, there is less evidence of this type of privacy leakage.
- If there are exact matches, IMS can reveal possible direct reproduction of training records.

Therefore, these metrics should be interpreted together rather than individually.

7.7.9 Strict Privacy Test

The three proxy metrics can be combined into a strict privacy test.

The synthetic dataset is considered private only if it passes: DCR_{test}, quad NNDR_{test}, quad IMS_{test} all together.

This provides a stricter evaluation because each metric captures a different aspect of potential privacy risk.

7.7.10 Continuous DCR

DCR can also be used as a continuous privacy measure by calculating the mean distance between synthetic records and their closest training records

A higher mean DCR indicates that synthetic records are farther from the training data and is therefore assumed to indicate better privacy.

However, DCR is a proxy measure and should not be treated as a formal privacy guarantee.

7.7.11 Results from Our Experiment

In our experiment:

overline{DCR}_{train}=0.0583
overline{DCR}_{nonmember}=0.0956

The training DCR is lower than the non-member DCR, meaning that synthetic records are, on average, closer to the training data.

The 5th percentiles show the same pattern:

DCR_{train}^{5%}=0.0277
DCR_{nonmember}^{5%}=0.0469
This provides a potential indication of greater similarity between synthetic and training records.

For IMS:

IMS_{train}=0,
qquad IMS_{nonmember}=0

Therefore, no exact matches were found in either dataset. This means there is no evidence of exact record reproduction, although it does not rule out other forms of privacy leakage.

These proxy results can therefore be complemented with membership inference attacks, which directly test whether an attacker can distinguish training members from non-members.

Ref :
1. Yao Z, Krčo N, Ganev G, de Montjoye YA. The dcr delusion: Measuring the privacy risk of synthetic data. InEuropean Symposium on Research in Computer Security 2025 Sep 22 (pp. 469-487). Cham: Springer Nature Switzerland.